

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019

Plant disorder precognition by image based pattern recognition

Varalakshmi P, Aravindkumar S*

^aANNA University – MIT, Chrompet, Chennai, 600 044, INDIA

Abstract

Proliferating population in the world has ushered a lot of burden on resources in agriculture. In order to sustain the population and the economy, it is very important to obtain maximum yield from crops. Diseases in plants are the chief cause of damage in plants which leads to production loss in agriculture. Earlier disease detection is manual, where farmers refer guide books or use their experiences to recognize the diseases. This process is time-consuming and requires some knowledge for selection of pesticides. Capturing images of leaves and processing them with the help of an automated system to identify the disease infected leaves could be a better solution for farmers. This paper discuss about how image processing and machine learning is effectively used in detection of diseases in plants. Analysis is done with various classification techniques and the comparison is made between Active contour and Sobel edge detection techniques. Experimental results indicate that Multiclass Support Vector Machines combined with Active contour edge detection gives better results compared to other methods for disease classification.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Edge detection; Classification; Active contour; Sobel operator; Multiclass SVM;

1. INTRODUCTION

Radical reduction in agricultural production and food security is due to the increase of plant disease. Since precision agriculture can provide a good growing environment for crops has been used widely nowadays. It also provides appropriate conditions for numerous microbes. Hence in the protected environment disease incidence and disease severity can be increased. Earlier methods to identify plant diseases and for diagnosing it, has mainly based on naked-eye study by the experts in field of microbes detection in laboratory. If personnel not experienced, then this

* Corresponding author. Tel.: 9942544146

E-mail address: saravind123@gmail.com

way will give an inaccurate results in additional with time consuming. Moreover, some expert systems can be used for identification and diagnosis of plant diseases. Most of the systems require disease-related information as input data, but the information provided by the receivers cannot satisfy the requirements since they do not have the necessary disease information. Therefore, the systems cannot give accurate outputs. This leads to excess use or misuse of fertilizers or pesticides and causes a big risk to food security.

Computer system broadly helped in data management, collection and analysis in the study on plant pathology. Using relevant methods recognition of the diseases and automatic classification of disease severity can be achieved. As a result, users can acquire associated disease information in time and precisely. Automation of disease detection by continuous monitoring of the crops, with image processing techniques can facilitate timely disease control which results in higher yield, quality of the crops is improved, and the quantity of the pesticides used will be decreased massively. The key objective of this work is to visualize the application of image processing and classification techniques in plant disease detection. Various methods such as Naïve Bayes, Decision tree, K-Nearest Neighbor, Support Vector Machines, Discriminant analysis, logistic regression, and Ensemble classifiers are compared and it is clear that SVM gives better results. Comparison is also made between Sobel and Active contour edge detection techniques where the result shows that Active contour detects the edge accurately than the Sobel. Implementation results reveals that the proposed approach Multiclass Support vector machine combined with Active contour edge detection surpasses other techniques.

2. LITERATURE SURVEY

Antoine Vacavant, Manuel Grand-Brochier, Camille Kurtz, Guillaume Cerutti, Laure Tougne, Jonathan Weber [1] compared a segmentation algorithms that are practical to the natural images for the extraction of tree leaves. The results presented in this paper show that the Guided Active Contour gives better results for all criteria that are considered. Z. Sarayloo and D. Asemani [2] modeled a classifier using pattern recognition techniques for detecting fungal diseases in wheat plants. For wheat disease classification a radial basis function is used. Singh, Vijai, and A. K. Misra[3] presents an algorithm for image segmentation technique by doing survey on different disease classification technique. R. Kaur and S. S. Kang [4] focused on detecting and classifying diseases in plants using the image processing techniques. In [5], N. Israni, A. Savla, P. Dhawan discussed different algorithms relating to classification techniques of data mining and finds the best one predicting the yield. For attaining diagnosis of diseases in plants automatically and to increase accuracy in recognizing plant diseases, H. Wang, X. Li, G. Li and Z. Ma [6] used 2 varieties of grape and wheat diseases as research objects. The work revealed that disease identification can effectively be achieved using BP networks. A. Bechar, N. Schor, T. Ignat, Y. Elad, A. Dombrovsky, and S. Berman [7] presented a detection system which is robotic, for the detection of Powdery mildew. This technique is also used in detecting Tomato spotted wilt virus. P. Revathi, M. Hemalatha [8] proposed an advance computing technology to take better decisions in crop development process. Using mobile images of leaf, symptoms of cotton leaf spot are expressed.

In [9] B. S. Prajapati, V. K. Dabhi and H. B. Prajapati done a survey of various machine-learning and image processing algorithms that are used in detecting diseases in rice plant based on rice plant images. In [10] A. Manjula and G. Narsimha, proposed a technique named XCYPF which is extensible. In [11] A. Akhtar, A. Khanum, S. A. Khan and A. Shaukat compared the results of various machine learning algorithms to identify and classify disease patterns in plant leaf images. S. D. Khirade and A. B. Patil[12], discussed the methods for detecting diseases in plants using their leaf images. The image enhancement techniques are classified into frequency and spatial domain. In [13] S. S. Bedi, Rati Khandelwal shows spatial domain enhancement processing technique. In [14] M. Paul, S. K. Vishwakarma and A. Verma presented a system for predicting the category of the soil datasets using data mining techniques. S. Veenadhari, B. Misra and C. Singh [15] used a software tool named 'Crop Advisor' for predicting the climatic parameters influence on the crop yields. In [16] diseases were identified only for maize plant based on the pre-trained models like googlenet, cifar100 models for classification. Region growing method is used for feature extraction. Based on those features, the classification of plant will be made [17]. Strong correlated pixel is used in [18] for segmentation, genetic algorithm used for feature selection and for identifying a disease KNN and SVM classifier are used. A new methodology called Particle swarm based SVM (P-SVM) is used in [19] to achieve more accuracy based on initialization parameters of classification and segmentation, where K means algorithm is used for segmentation and SVM is used for classification.

On taking indication of above mentioned work, the classification techniques are considered and the work flow is followed in the proposed system.

3. PROPOSED SYSTEM

The objective is to diagnose disease type that has affected the plant. In order to do these, the plant leaf images are captured over a period of time with some particular time interval. The machine learning and data mining techniques play a vital role in this plant disease precognition system

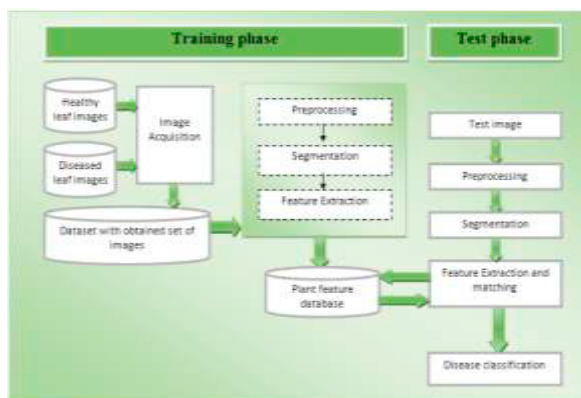


Fig. 1. Ensemble feature based classification Architecture

The work flow of the system is shown in Fig.1 The flow consists of training and testing phase. Under the training phase, the image acquisition block collects the set of healthy and diseased leaf images those are necessary to train the system for classification of diseases. The set of images has been separated as train and test sets. The training set of images is given to the classifier model to learn the system with various types of diseases. The accuracy of the classifier model is analyzed using the test set images. Using the images that are collected classifier model is created. These images are first preprocessed and the resulting images are being applied to the segmentation phase. The segmentation phase then fragments the diseased leaf into clusters. The detailed discussion of the preprocessing and segmentation phase is given in the subsequent sections. The features are then extracted from the user selected cluster which contains the Region of Interest (ROI) of the diseased part and stored in the feature database. All these processes happen in the training phase of the system. The data obtained in training phase is given to the classifiers to learn the system. Then the test set feature database is given to the system and the disease prediction is done. This phase analyzes the accuracy of the disease prediction. The testing phase images are pre-processed, the Region of Interest (ROI) is obtained from segmentation phase and the features are extracted. Finally, the classification system, diagnoses the disease that has affected the leaf. Several algorithms are been compared under various phases of the proposed work. The algorithms under each phase are explained below.

3.1 Pre – processing

For better results in following stages, image pre-processing is needed because dust, dewdrops, shadow effect may present on the object. Presence of all such misinterpretation can be reduced or isolated using noise removal filters.

3.1.1 Gaussian Filter

Gaussian blur or blurring an image is resulted by gaussian function. It is basically for reducing image noise. Filter the leaf image with isotropic gaussian smoothing kernels of increasing standard deviations. This filter use the same standard deviation along both dimensions.

3.1.2 Sharpening

Sharpening an image enhances the contrast between bright and dark regions to bring out features. In this method the chosen leaf image is passed to a high-pass filter to make it appear sharper. In order to enhance the edges, the sharpening method is applied to the filtered image obtained from Gaussian filter.

3.1.3 Contrast Enhancement

For image quality evaluation one of the important factors is contrast. The sharpened image is converted to L*a*b color space. The sharpening process happens then with the help of the filter radius value. The processed image is converted to RGB colors pace. It increases the image contrast by mapping the values of intensity to new values. The process flow is given below.

Input: Filtered RGB image from Gaussian

1. Convert RGB image to L*a*b color space
2. Initialize radius and amount to default values
3. Assign the filter radius to the ceil of radius * 3
4. Set filter size to thrice of filter radius + 1

5. Assign sharpen filter to 1 initially
6. Update sharpen filter to amount * sharpen filter
7. Perform the updating of sharpen filter by incrementing by 1
8. Convert L*a*b image to RGB

Output: Enhanced RGB image

3.2 Edge detection

Prior to the segmentation the edge detection algorithms have been applied on the images. The actual leaf part to be processed is extracted from the background using edge detection techniques. The edges have been detected with sobel and active contour edge detection mechanisms which are discussed in the following sections:

3.2.1 Active contour edge detection

Active contour is a dynamic edge detection technique. For this method to find the edges, initial mask is set up to the required size. The dynamic edge detection happens with the process of movement of the curvature from the initial mask towards the actual edges of the image. This method changes the curvature dimensions for each iteration and extracts the leaf from its background. The edge functional is based on the image gradient. One implementation of this is given in (1).

$$E_{\text{edge}} = -|\Delta I(x, y)|^2 \quad (1)$$

The pseudocode of the Active contour edge detection algorithm is given below.

Active contour edge detection algorithm

Input: Contrast enhanced RGB image

1. Initialize the mask to necessary size
2. Scale the input image to the required size
3. Convert mask to Phi values with maskToPhi()
4. For each iteration
 - a. Set idx as phi between the range 1.2 and -1.2
 - b. Segregate negative and postive phi values
 - c. Then calculate x and y as follows
 - d. $x = \text{sum}(I(\text{xpts})) / (\text{length}(\text{xpts}) + \text{eps});$
 - e. $y = \text{sum}(I(\text{ypts})) / (\text{length}(\text{ypts}) + \text{eps});$
 - f. Calculate $Z = (I(\text{idx}) - x)^2 - (I(\text{idx}) - y)^2;$
5. Get curvature from the obtained idx and phi
6. End

Output: Edge detected image

Ensemble model boosting algorithm

Input: Set of class before weight assigned

Initialize weight of each class to 0

For each classifier:

$w_i = \log 1 - \text{error}(M_i)$

$c = M_i(X)$

Add w_i to the weight for class c

End for

Return the class with largest weight

Output: Class with largest weight

3.2.2 Sobel edge detection

The Sobel method is used in computer vision and image processing technique, and also in creating an image emphasizing edges while used in edge detection algorithms.

3.3 Classification

The classification phase not only identifies the affected leaves but also detects the type of disease that has affected the leaf. The classification is done with different classifiers and the results are compared.

3.3.1 Support Vector Machine (SVM)

In case of support vector machines, p - dimensional vector is used for viewing a data point. To identify whether those points can be separated with a $(p-1)$ dimensional hyperplane is the main objective. In the proposed system, the multiclass SVM uses $p-1$ hyperplanes in order to separate the classes of the diseases that the leaf has been affected with.

3.3.2 Decision tree

In this model, the tree structure is used to build a classification model. It splits the entire input dataset into minor sets and a tree is gradually developed. The final outcome is a tree with decision nodes which has the features extracted from the leaf and leaf nodes that specifies the type of disease. Finding attribute that returns the highest information gain is a decision tree construction. The information gain is calculated with the following formula in equation (2).

$$\text{Gain}(A,B) = \text{Entropy}(A) - \text{Entropy}(A,B) \quad (2)$$

3.3.3 Naive Bayes

Bayes theorem calculates the posterior probability, $P(c|x)$, from $P(x)$, $P(c)$, and $P(x|c)$. This classifier has an assumption that the effect of a feature (x) extracted from an image on a given disease class (c) is independent of other features. This relation is called class conditional independence. The probability $P(C|x)$ is calculated as given in equation (3).

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \quad (3)$$

3.3.4 Discriminant analysis

Discriminant function analysis is used for predicting a dependent variable by one or more independent variables. Specifically, discriminant analysis predicts a classification (X) variable based on known continuous responses (Y). The disease prediction done with this classifier is better than many other classifiers as discussed in the results section.

3.3.5 K-Nearest Neighbor

In this model, a class is assigned by votes of its neighboring predictions. The resultant class allocated to the leaf node is the most common amongst its K nearest neighbors measured by the distance criteria. The optimal value for K could be best chosen by first inspecting the data. Typically, the optimal K for most datasets has been between 3 and 10. The distance functions those are used by the KNN algorithm namely Euclidean distance, Manhattan distance and Minkowski distance have been discussed in the following (4), (5) and (6) respectively.

$$\text{Euclidean distance} = \sqrt{\sum_{i=1}^k (x_i - y_i)^2} \quad (4)$$

$$\text{Manhattan distance} = \sum_{i=1}^k |x_i - y_i| \quad (5)$$

$$\text{Minkowski distance} = (\sum_{i=1}^k (|x_i - y_i|^q))^{1/q} \quad (6)$$

3.3.6 Ensemble model

Ensemble method used in here is Boosting. This is an iterative model that adjusts the weight value of a prediction based on the previous classification. If a prediction was misclassified, this model tries to increase the weight value of the observation. Boosting often reduces the bias error and builds very good predictive models. Ensemble model boosting algorithm is explained below.

3.3.7 Logistic regression

Logistic regression can be used only when the target variable is either a categorical variable that has exactly two categories or a continuous variable that has values in the range 0.0 to 1.0.



Fig 2. Sample dataset

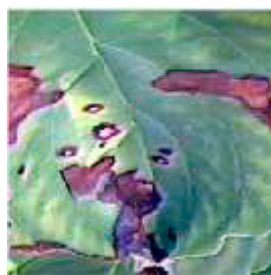


Fig 3. Image obtained after pre-processing

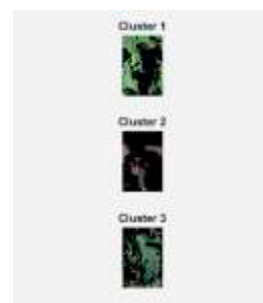


Fig 4. Segmented clusters

4. EXPERIMENTS

The images of healthy and diseased leaves are collected from the PlantVillage website [20] and the database is formed. The diseases comprise of the categories namely Alternaria Alternata, Anthracnose, Bacterial Blight, Black spot, Cercospora Leaf spot of Corn, Potato and Tomato plants. The dataset does not include any overlapping or occlusion. The image samples of all the above mentioned diseases and the healthy leaves are shown in Fig. 2 of potato plant. The features have been extracted from the collected set of images. The implementation of the proposed system which includes various methods is executed in Matlab. Implementation of the proposed work led to the visualization of the following results. The experimental study and its results mainly will focus on comparison of the classification algorithms. The objective of the proposed system is to balance the results of various classification methods mentioned in Chapter 3. The input image is taken from the test set of images.

Fig.3. denotes the output image obtained after application of the Gaussian filter, image sharpening and the contrast enhancement techniques. These sets of processing and the techniques are done under image pre-processing phase in which the image is enhanced and made suitable for further processing. After performing the pre-processing of the images, the image represented in Fig. 3 is passed to the Active contour and Sobel edge detection techniques. The result of edge detection algorithm is given as input to segmentation algorithm. From these techniques the leaf is extracted from its background and the actual leaf is obtained. The K-means clustering technique then clusters the leaf parts into 3 segments as mentioned in the Fig.4. The user is requested to select the ROI (i.e) the cluster having the diseased part of the leaf. The proposed system compares the performance of various edge detection techniques and several classifiers. The graphs mentioned in the next section show the comparison results. The parameters considered for performance evaluation are Specificity, Sensitivity, Accuracy, positive predictive value and negative predictive value. The gray scale image of the interested ROI is displayed in the Fig.5. The dynamic edge detection of Active contour method is shown in Fig.6. The parameters considered for performance evaluation are Specificity, Sensitivity, Accuracy, positive predictive value and negative predictive value.

- **Accuracy:** Correctly Classified Samples / Classified Samples
- **Sensitivity:** Positive Samples that are Correctly Classified / True Positive Samples
- **Specificity:** Negative Samples that are Correctly Classified / True Negative Samples
- **Positive predictive value (PPV):** Positive Samples that are correctly classified / Positive Classified Samples
- **Negative predictive value (NPV):** Negative Samples that are Correctly Classified / Negative Classified Samples

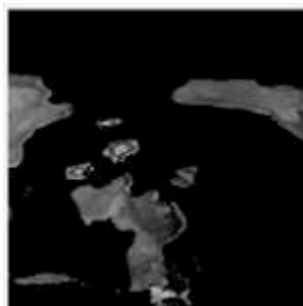


Fig 5. Grayscale image

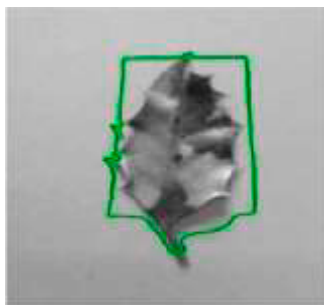


Fig 6. Active Contour Edge Detection

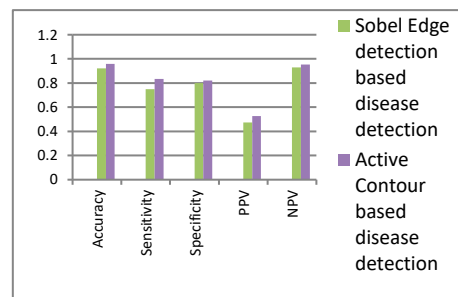


Fig 7. Comparison of edge detection algorithms

5. RESULTS AND DISCUSSION

The graph in Fig.7 shows that the Active Contour edge detection method shows comparatively better performance than the sobel edge detection method. The classifiers considered for comparison of performance in the proposed disease prediction system includes the following: SVM, Decision tree, K - Nearest Neighbor, Naïve Bayes, Logistic regression, Ensemble and Discriminant Analysis. Figure 8 shows the result of sobel edge detection on the leaf of potato plant. The graph in Fig.9 shows the Accuracy of the classifiers in the proposed disease prediction. From the results pictured in the graph, it is evident that the Multiclass SVM method has better performance than the other classifiers. The SVM method works with hyper plane and hence it provides the best classification compared to other techniques. The graph in Fig.10 shows the Sensitivity of the classifiers in the proposed disease prediction. From the results pictured in the graph, it is shown that the Discriminant Analysis classifier has better sensitivity than the other classifiers. Whereas the value is near to the SVM classifier. The graph in Fig. 11 shows the Specificity of the classifiers in the proposed disease prediction. From the results pictured in the graph, it is shown that the SVM classifier has better sensitivity than the other classifiers. On an average all the classifiers listed in the graph have similar values of specificity.



Fig 8. Result of sobel edge detection

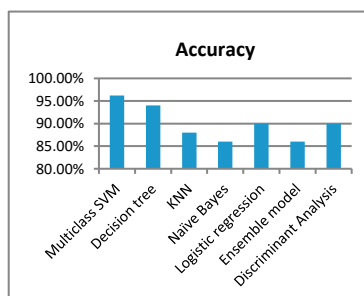


Fig 9. Comparison of Accuracy of classifiers

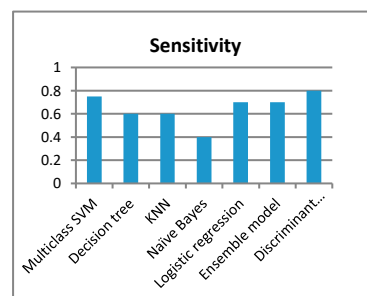


Fig 10. Comparison of Sensitivity of classifiers

The graph in Fig.12 shows the Positive predictive value of the classifiers in the proposed disease prediction. From the results pictured in the graph, it is shown that the Decision tree, KNN and ensemble classifiers have similar values and they show better values than the other classifiers. The graph in Fig.13 shows the Negative predictive value of the classifiers in the proposed disease prediction. From the results pictured in the graph, it is shown that the Discriminant analysis has the best value when compared to the other classifiers.

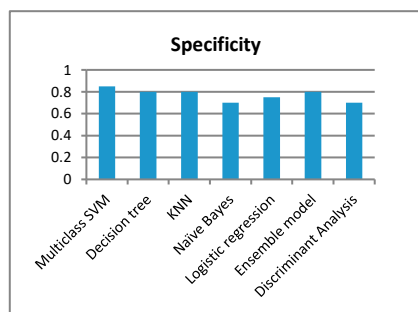


Fig 11. Comparison of Specificity of classifiers

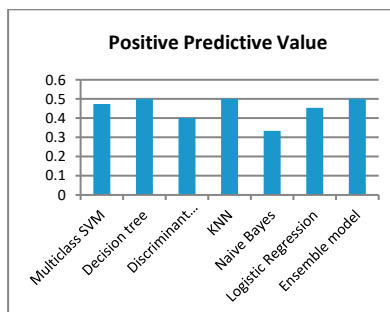


Fig 12. Comparison of PPV of classifiers

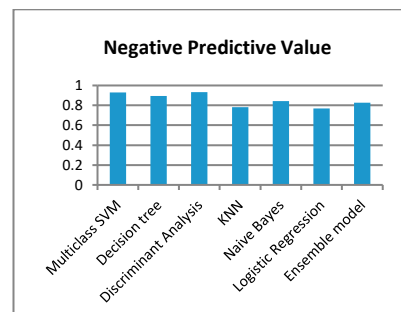


Fig 13. Comparison of NPV of classifiers

6. CONCLUSION AND FUTURE WORK

Plant diseases leads to incredible quantity of loss in agriculture if proper attention is not given. An automated system can be a solution for the stated problem. Hence to overcome, the system has been proposed using various methods. In this system, some of the machine learning and classification technique used in detecting the disease have been reviewed and summarized learning. For achieving visually acceptable images, the images are modified accordingly using the approaches offered by image enhancement algorithms. The review of Image enhancement techniques such as Gaussian filter, image sharpening and image contrast enhancement has been successfully accomplished. The diseased region extraction from the leaf image is the significant step, for which the edge detection techniques such as Active Contour and Sobel edge detection methods has been considered and the results are analyzed. The disease classification phase performs the classification of the diseases and also identifies the type of disease that has affected the leaf. For classification, several supervised techniques like SVM, Decision tree, Naive Bayes, Discriminant Analysis, KNN and Ensemble models are used and the comparison results are discussed. From the discussion it is clear that, the SVM classifier has comparatively higher accuracy than the other classifiers discussed in the system.

References

- [1] Grand-Brochier, Manuel, Antoine Vacavant, Guillaume Cerutti, Camille Kurtz, Jonathan Weber, and Laure Tougne. (2015) Tree leaves extraction in natural images: Comparative study of preprocessing tools and segmentation methods." *IEEE Transactions on Image Processing* **24** (5): 1549-1560.
- [2] Sarayloo, Zahra, and Davud Asemani. (2015) "Designing a classifier for automatic detection of fungal diseases in wheat plant: By pattern recognition techniques." *In Electrical Engineering (ICEE)* 1193–1197.
- [3] Singh, Vijai, and A. K. Misra. (2015) "Detection of plant leaf diseases using image segmentation and soft computing techniques." *Information Processing in Agriculture* **4** (1): 41–49.
- [4] Kaur, Rajleen, and Sandeep Singh Kang (2015) "An enhancement in classifier support vector machine to improve plant disease detection." *Innovation and Technology in Education (MITE)*, 135–140.
- [5] Savla, Anshal, Nivedita Israni, Parul Dhawan, Alisha Mandholia, Himtanaya Bhadada, and Sanya Bhardwaj. (2015) "Survey of classification algorithms for formulating yield prediction accuracy in precision agriculture." *Innovations in Information, Embedded and Communication Systems (ICIIECS)*, 1–7.
- [6] Wang, Haiguang, Guanlin Li, Zhanhong Ma, and Xiaolong Li. (2012) "Image recognition of plant diseases based on backpropagation networks." *5th International Congress on Image and Signal Processing (CISP)*, 894-900.
- [7] Schor, Noa, Avital Bechar, Timea Ignat, Aviv Dombrovsky, Yigal Elad, and Sigal Berman. (2016) "Robotic disease detection in greenhouses: combined detection of powdery mildew and tomato spotted wilt virus." *IEEE Robotics and Automation Letters* **1**, 354-360.
- [8] Revathi, P., and M. Hemalatha. (2015) "Classification of cotton leaf spot diseases using image processing edge detection techniques." *In Emerging Trends in Science, Engineering and Technology (INCOSSET)*, 169–173.
- [9] Shah, Jitesh P., Harshadkumar B. Prajapati, and Vipul K. Dabhi. (2016) A survey on detection and classification of rice plant diseases." *IEEE International Conference on Current Trends in Advanced Computing (ICCTAC)*, 1–8.

- [10] Manjula, Aakunuri, and G. Narsimha. (2015) “XCYPF: A flexible and extensible framework for agricultural Crop Yield Prediction.” *IEEE 9th International Conference on Intelligent Systems and Control (ISCO)*, 1 –5.
- [11] Akhtar, Asma, Aasia Khanum, Shoab A. Khan, and Arslan Shaukat (2013) “Automated Plant Disease Analysis (APDA): Performance Comparison of Machine Learning Techniques.” *Frontiers of Information Technology(FIT)*, 60–65.
- [12] Khirade, Sachin D., and A. B. Patil. (2015) “Plant Disease Detection Using Image Processing.”, *International Conference on Computing Communication Control and Automation (ICCUBE)*, 768-771.
- [13] Bedi, S. S., and Rati Khandelwal. (2015) “Various image enhancement techniques-a critical review.”, *International Journal of Advanced Research in Computer and Communication Engineering*.
- [14] Paul, Monali, Santosh K. Vishwakarma, and Ashok Verma. (2015) “Analysis of Soil Behaviour and Prediction of Crop Yield Using Data Mining Approach.” *International Conference on Computational Intelligence and Communication Networks(CICN)* , 766-771.
- [15] Veenadhari, S., Bharat Misra, and C. D. Singh. (2014) “Machine learning approach for forecasting crop yield based on climatic parameters.” *In Computer Communication and Informatics (ICCCI)*, 1–5.
- [16] Xihai Zhang, Yue Qiao, Fanfeng Meng, Chengguo Fan and Mingming Zhang. (2018) “Identification of Maize Leaf Diseases Using Improved Deep Convolutional Neural Networks.”, *IEEE Access*, 30370 -30377.
- [17] Siddharth Singh Chouhan, Ajay Kaul, Uday Pratap Singh and Sanjeev Jain. (2018) “Bacterial Foraging Optimization Based Radial Basis Function Neural Network (BRBFNN) for Identification and Classification of Plant Leaf Diseases: An Automatic Approach Towards Plant Pathology.”, *IEEE Access – translation and mining*, 8852 -8863.
- [18] Muhammad Attique Khan, , M Ikram Ullah Lal, Muhammad Sharif and Kashif Javed. (2019) “An Optimized Method for Segmentation and Classification of Apple Diseases Based on Strong Correlation and Genetic Algorithm Based Feature Selection.”, *IEEE Access – translation and mining* , 46261 -46277.
- [19] Vippon Preet Kour and Sakshi Arora. (2019) “Particle Swarm Optimization Based Support Vector Machine (P-SVM) for the Segmentation and Classification of Plants.”, *IEEE Access – translation and mining*, 8852 -8863.
- [20] David. P. Hughes, Marcel Salathe, “An open access repository of images on plant health to enable the development of mobile disease diagnostics”, arXiv:1511.08060, 2015